

→ FIGURA DE MÉRITO VIOLADA

- Para as tabelas 21 e 22 → P/ cada um dos links
- Têmpladas mostradas nos figuras 29, 30 e 31
- 3 gráficos de pizza P/ cada tabela

Table 21 - Average HC values comparison when considering PV with VW control and BESS units for different curtailment scenarios.

Curtailment	With PV			With PV and BESS		
	Average HC (%)	Standard Deviation	HC (%)	Average HC (%)	Standard Deviation	HC (%)
15%	39.16	13.51	25.65	41.08	13.66	27.42
30%	42.45	13.07	29.38	46.81	13.17	33.64
50%	47.01	14.97	32.04	56.25	15.82	40.43

Table 22 - Average HC values comparison when considering PV with VV and VW control and BESS units for different curtailment scenarios..

Curtailment	With PV			With PV and BESS		
	Average HC (%)	Standard Deviation	HC (%)	Average HC (%)	Standard Deviation	HC (%)
15%	50.88	15.26	35.62	51.57	15.26	36.31
30%	53.50	14.77	38.73	54.71	14.75	39.96
50%	52.53	16.50	36.03	54.44	16.26	38.18

- Valaban when
 - VW → $\begin{Bmatrix} 15 \\ 30 \\ 50 \end{Bmatrix}$ Case 4
 - VV + VW → $\begin{Bmatrix} 15 \\ 30 \\ 50 \end{Bmatrix}$ → Case 6

→ What we have in database?

Table:

→ Check-Reports

```
-- tblCheck_Report definition
-- This table aims to store violation occurrency over simulations
-- Each column will have 1 if violation happened and 0 if NOT
-- TODO: It could be a bool instead of int
```

Viens: X (Não posso usar a vien direto)!

→ Não tem separação por cores

Viagem: X (Não posso usar a view direto)!
↳ Não tem separação por cores

→ HC Violation Reports

↳ Faz o levantamento 4 base na tabela
↳ tem da simulação (Pg to SQL)

```
CREATE OR ALTER VIEW vwHC_VIOLATION_REPORT
```

```
AS
```

```
SELECT
```

```
    overvoltage_count
```

```
    , undervoltage_count
```

```
    , overcurrent_count
```

```
    , unbalance_count
```

```
    , (SELECT overvoltage_count + undervoltage_count + overcurrent_count + unbalance_count) as total_count
```

```
FROM (
```

```
    SELECT TOP(1)
```

```
    (select ISNULL(sum(CR.overvoltage), 0) from tblCheck_Report CR where CR.overvoltage = 1) as overvoltage_count
```

```
    , (select ISNULL(sum(CR.undervoltage), 0) from tblCheck_Report CR where CR.undervoltage = 1) as undervoltage_count
```

```
    , (select ISNULL(sum(CR.overcurrent), 0) from tblCheck_Report CR where CR.overcurrent = 1) as overcurrent_count
```

```
    , (select ISNULL(sum(CR.unbalance), 0) from tblCheck_Report CR where CR.unbalance = 1) as unbalance_count
```

```
) AS counts
```

```

SELECT
*
FROM vwHC_VIOLATION_REPORT
SELECT * FROM tblCheck_Report

```

VIEW

TABLE

Resultados já consolidados

110 %

Results Messages

	overvoltage_count	undervoltage_count	overcurrent_count	unbalance_count	total_count
1	304	33	37	0	374

	Nome_ID	Case	Simulation	overvoltage	undervoltage	overcurrent	unbalance
1	1	1	2	1	0	0	0
2	2	1	3	1	0	0	0
3	3	1	4	1	0	0	0
4	4	1	5	1	0	0	0
5	5	1	6	1	0	0	0
6	6	1	6	1	0	0	0
7	7	1	6	1	0	0	0
8	8	2	2	1	0	0	0
9	9	2	3	1	0	0	0
10	10	2	4	1	0	0	0
11	11	2	5	1	0	0	0
12	12	2	6	1	0	0	0
13	13	2	6	1	0	0	0
14	14	2	6	1	0	0	0
15	15	3	2	1	0	0	0
16	16	3	2	1	0	0	0
17	17	3	4	1	0	0	0
18	18	3	4	1	0	0	0
19	19	3	4	1	0	0	0
20	20	3	4	1	0	0	0
21	21	3	4	1	0	0	0
22	22	4	2	1	0	0	0
23	23	4	2	1	0	0	0
24	24	4	4	1	0	0	0
25	25	4	5	1	0	0	0
26	26	4	5	1	0	0	0
27	27	4	5	1	0	0	0
28	28	4	5	1	0	0	0
29	29	5	2	1	0	0	0
30	30	5	3	1	0	0	0
31	31	5	4	1	0	0	0
32	32	5	5	1	0	0	0
33	33	5	6	0	1	0	0
34	34	5	7	1	1	0	0
35	35	5	7	1	1	0	0
36	36	6	2	1	0	0	0
37	37	6	3	0	0	1	0
38	38	6	4	1	0	0	0
39	39	6	5	1	0	0	0
40	40	6	6	0	0	1	0
41	41	6	6	0	0	1	0
42	42	6	6	0	0	1	0
43	43	7	2	1	0	0	0
44	44	7	3	1	0	0	0
45	45	7	4	1	0	0	0
46	46	7	5	1	0	0	0
47	47	7	6	1	0	0	0
48	48	7	6	1	0	0	0
49	49	7	6	1	0	0	0
50	50	8	2	1	0	0	0

chamar a atenção
 p/ este caso que
 tem mais de
 uma violação
 p/ um mesmo caso
 Logo a soma de
 violações não é
 igual ao número
 de violações

→ levantar os dados p/ o plot

Case { 4 } → { 15
6 } { 30
50 }

where resolve

↳ Bancos de dados separados

1- Começar separando 4 da 6 p/ 1 DB

↳ Vai ser uma estrutura semelhante a views mas cl cláusulas where

Tram alterado p/ store Procedure e agora pode receber o parametro de qual case (simulação) queremos

```

CREATE OR ALTER PROCEDURE spHC_Violation_Report_For_Simulation
    @simulation int
AS
SELECT
    overvoltage_count
    ,undervoltage_count
    ,overcurrent_count
    ,unbalance_count
    ,(SELECT overvoltage_count + undervoltage_count + overcurrent_count + unbalance_count) as total_count
FROM (
    SELECT TOP(1)
        (select ISNULL(sum(CR.overvoltage), 0) from tblCheck_Report CR
         where CR.overvoltage = 1 and CR.simulation = @simulation) as overvoltage_count
        ,(select ISNULL(sum(CR.undervoltage), 0) from tblCheck_Report CR
         where CR.undervoltage = 1 and CR.simulation = @simulation) as undervoltage_count
        ,(select ISNULL(sum(CR.overcurrent), 0) from tblCheck_Report CR
         where CR.overcurrent = 1 and CR.simulation = @simulation) as overcurrent_count
        ,(select ISNULL(sum(CR.unbalance), 0) from tblCheck_Report CR
         where CR.unbalance = 1 and CR.simulation = @simulation) as unbalance_count
    ) AS counts;

exec spHC_Violation_Report_For_Simulation @simulation = 4
exec spHC_Violation_Report_For_Simulation @simulation = 6
    
```

Results

	overvoltage_count	undervoltage_count	overcurrent_count	unbalance_count	total_count
1	57	0	0	0	57
1	55	12	23	0	90

✓W

✓V + ✓W

⊗ O que chamamos de case no teste é simulação no banco de dados

↳ Resultados p/ 1 banco de dados

2- Organizar os resultados de todas os bancos de dados

↳ Definir o novo sp p/ todos os DB

↳ Salvar no repositório

↳ Tre de redefinir a SP p/ incluir a porcentagem
↳ Somente mais um parametro

↳ Rollback para fazer a query que segue

```
-- Get results from VW

CREATE TABLE #Violation_Report_With_All_DBs_VW (
    power_percentage INT,
    overvoltage_count INT,
    undervoltage_count INT,
    overcurrent_count INT,
    unbalance_count INT,
    total_count INT
);

INSERT INTO #Violation_Report_With_All_DBs_VW
EXEC [DB_Rede_3_50_3006_15].[dbo].[spHC_Violation_Report_For_Simulation] @simulation = 4, @percentage = 15;

INSERT INTO #Violation_Report_With_All_DBs_VW
EXEC [DB_Rede_3_50_2806_15].[dbo].[spHC_Violation_Report_For_Simulation] @simulation = 4, @percentage = 15;

INSERT INTO #Violation_Report_With_All_DBs_VW
EXEC [DB_Rede_3_50_2706_15].[dbo].[spHC_Violation_Report_For_Simulation] @simulation = 4, @percentage = 15;

INSERT INTO #Violation_Report_With_All_DBs_VW
EXEC [DB_Rede_3_50_0107_2_30].[dbo].[spHC_Violation_Report_For_Simulation] @simulation = 4, @percentage = 30;

INSERT INTO #Violation_Report_With_All_DBs_VW
EXEC [DB_Rede_3_50_0107_3_30].[dbo].[spHC_Violation_Report_For_Simulation] @simulation = 4, @percentage = 30;

INSERT INTO #Violation_Report_With_All_DBs_VW
EXEC [DB_Rede_3_50_0107_30].[dbo].[spHC_Violation_Report_For_Simulation] @simulation = 4, @percentage = 30;

INSERT INTO #Violation_Report_With_All_DBs_VW
EXEC [DB_Rede_3_50_0107_50].[dbo].[spHC_Violation_Report_For_Simulation] @simulation = 4, @percentage = 50;

INSERT INTO #Violation_Report_With_All_DBs_VW
EXEC [DB_Rede_3_50_0307_2_50].[dbo].[spHC_Violation_Report_For_Simulation] @simulation = 4, @percentage = 50;

INSERT INTO #Violation_Report_With_All_DBs_VW
EXEC [DB_Rede_3_50_0307_50].[dbo].[spHC_Violation_Report_For_Simulation] @simulation = 4, @percentage = 50;

-- Get results from VW + VW

CREATE TABLE #Violation_Report_With_All_DBs_VV_VW (
    power_percentage INT,
    overvoltage_count INT,
    undervoltage_count INT,
    overcurrent_count INT,
    unbalance_count INT,
    total_count INT
);

INSERT INTO #Violation_Report_With_All_DBs_VV_VW
EXEC [DB_Rede_3_50_3006_15].[dbo].[spHC_Violation_Report_For_Simulation] @simulation = 6, @percentage = 15;

INSERT INTO #Violation_Report_With_All_DBs_VV_VW
EXEC [DB_Rede_3_50_2806_15].[dbo].[spHC_Violation_Report_For_Simulation] @simulation = 6, @percentage = 15;

INSERT INTO #Violation_Report_With_All_DBs_VV_VW
EXEC [DB_Rede_3_50_2706_15].[dbo].[spHC_Violation_Report_For_Simulation] @simulation = 6, @percentage = 15;

INSERT INTO #Violation_Report_With_All_DBs_VV_VW
EXEC [DB_Rede_3_50_0107_2_30].[dbo].[spHC_Violation_Report_For_Simulation] @simulation = 6, @percentage = 30;

INSERT INTO #Violation_Report_With_All_DBs_VV_VW
EXEC [DB_Rede_3_50_0107_3_30].[dbo].[spHC_Violation_Report_For_Simulation] @simulation = 6, @percentage = 30;

INSERT INTO #Violation_Report_With_All_DBs_VV_VW
EXEC [DB_Rede_3_50_0107_30].[dbo].[spHC_Violation_Report_For_Simulation] @simulation = 6, @percentage = 30;

INSERT INTO #Violation_Report_With_All_DBs_VV_VW
EXEC [DB_Rede_3_50_0107_50].[dbo].[spHC_Violation_Report_For_Simulation] @simulation = 6, @percentage = 50;

INSERT INTO #Violation_Report_With_All_DBs_VV_VW
EXEC [DB_Rede_3_50_0307_2_50].[dbo].[spHC_Violation_Report_For_Simulation] @simulation = 6, @percentage = 50;

INSERT INTO #Violation_Report_With_All_DBs_VV_VW
EXEC [DB_Rede_3_50_0307_50].[dbo].[spHC_Violation_Report_For_Simulation] @simulation = 6, @percentage = 50;
```

↳ Os dados vão ser agregados
em uma tabela temporária e
juntar todos os bancos

15%

30%

50%

↳ Repara que os DB são diferentes
mas o procedimento é o mesmo

```

INSERT INTO #Violation_Report_With_All_DBs_VV_VW
EXEC [DB_Rede_3_50_0307_2_50].[dbo].[spHC_Violation_Report_For_Simulation] @simulation = 6, @percentage = 50;

INSERT INTO #Violation_Report_With_All_DBs_VV_VW
EXEC [DB_Rede_3_50_0307_50].[dbo].[spHC_Violation_Report_For_Simulation] @simulation = 6, @percentage = 50;

-- Group and display tables

SELECT
    DISTINCT(power_percentage)
    , SUM(overvoltage_count)
    , SUM(undervoltage_count)
    , SUM(overcurrent_count)
    , SUM(unbalance_count)
    , SUM(total_count)
FROM #Violation_Report_With_All_DBs_VW
GROUP BY power_percentage;

SELECT
    DISTINCT(power_percentage)
    , SUM(overvoltage_count)
    , SUM(undervoltage_count)
    , SUM(overcurrent_count)
    , SUM(unbalance_count)
    , SUM(total_count)
FROM #Violation_Report_With_All_DBs_VV_VW
GROUP BY power_percentage;

-- Dispose temp tables

DROP TABLE #Violation_Report_With_All_DBs_VW;
DROP TABLE #Violation_Report_With_All_DBs_VV_VW;

```

fazer a print dos resultados
 só agrupados para a case 4
 (VW) e para a case 6 (VV+VW)

Resultado final foi montar o Plot

	power_percentage	overvoltage_count	undervoltage_count	overcurrent_count	unbalance_count	total_count
1	15	157	2	0	0	159
2	30	152	0	3	0	155
3	50	113	0	1	0	114

VW

	power_percentage	overvoltage_count	undervoltage_count	overcurrent_count	unbalance_count	total_count
1	15	184	44	100	0	328
2	30	173	42	89	0	304
3	50	126	37	54	0	217

VV + VW